

Studio 9

Assignment 3 Framework Continued

Objectives

To continue to build the framework for assignment 3.

Exercise 1 (Studio Handin Exercise)

You will be extending the Assignment 3 framework from Studio 9 to allow the application to store any number of data sets and display each data set individually.

1. Download the template project and open it in Processing. Open the class diagram. Open your Studio 9 sketch and save it as Studio 10. Then copy all of the code from the sketch tab of the template project and replace all of the code in your studio 10 sketch tab.

In the class diagram changes to the DataSet class are in red. Also the button class is not included in the diagram as students may have included buttons in a variety of ways. Therefore the button list has been explicitly stated in the GraphManager class instead.

2. Implement the changes to the DataSet class using the notes below:

Modify the constructor so that the name field is set to the filename without the .csv extension. Hint: use the split method and split on the '.' character.

Create the getName() method that returns back the name of the data set.
3. Implement the GraphManager class using the notes below:

The getFileNames() method is a private method that will return back an array of filenames that are stored in the folder that is passed to the method. Hint: search how to do this on the processing website. Create this method first before implementing the constructor.

The constructor will create a new array list object for the list of data sets and the list of buttons. It will create an array which holds color values and initialise the array when it is declared with

a series of colour values of your choosing. It will set the bottom field to the value passed in. It will then get all the filenames stored in the folder that is passed into the constructor. Then it will go through all the filenames and create a new data set for each file using a colour value from the array as the colour of the data set.

The `plotDataSet()` method will check that the index value passed in is valid and then plot that data set using the bottom field value as the bottom of the graph.

Test your graph manager by uncommenting the appropriate code in the sketch tab and make sure you can plot the various data sets by pressing the number keys.

4. Extend the constructor so that after adding all the data sets, it creates a new button for each dataset and stores the button in the button list. Use the name of the current data set as the caption for the button and choose an appropriate width, height and colour for the button and there should be a small gap between each button (see the screen shots at the end of the pdf).

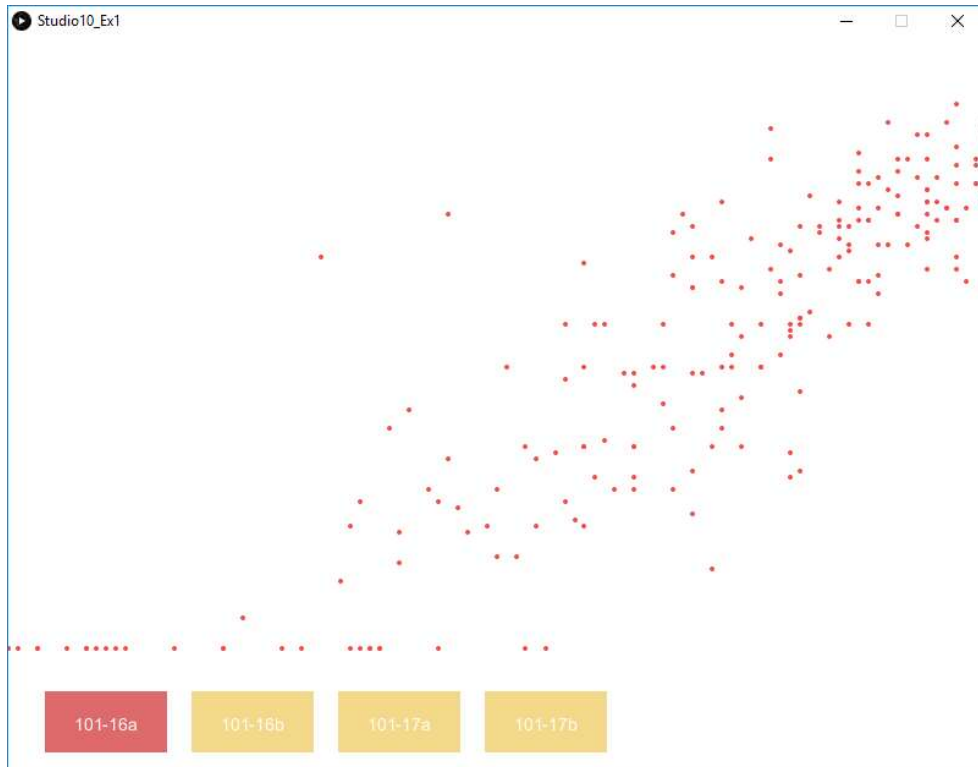
5. Create the `drawUI()` method that draws all the buttons to the sketch window.

Test your sketch to see if the buttons are drawn correctly by uncommenting the appropriate code in the sketch tab.

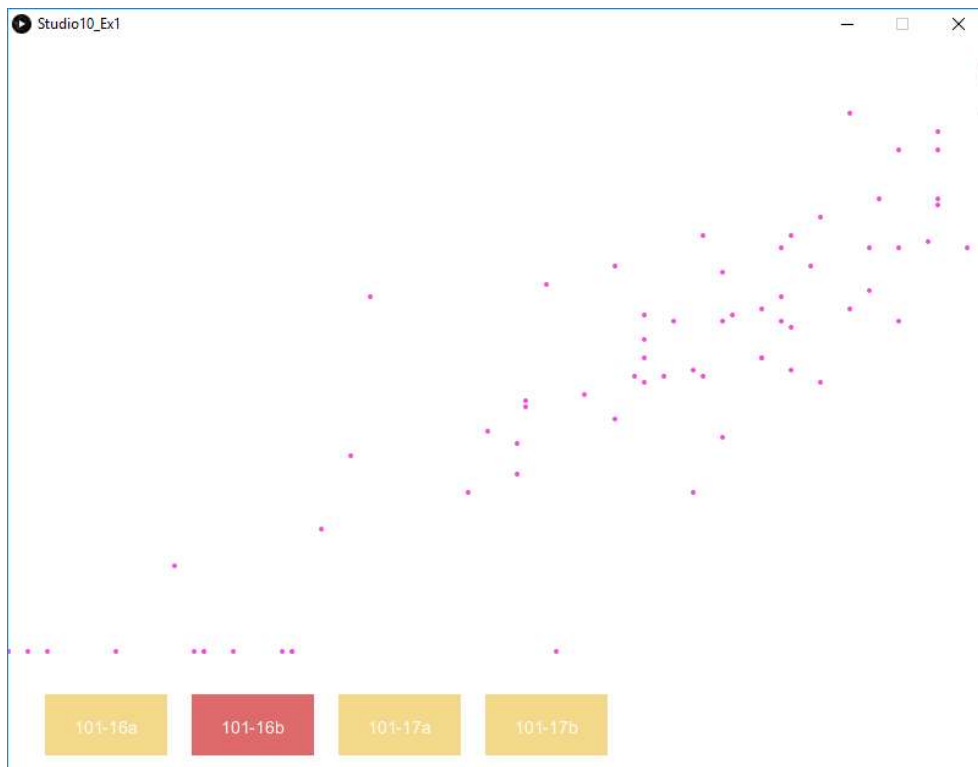
6. Create the `checkButtonClick()` method that returns back the index position of the button that has been clicked on or returns back -1 if no button has been clicked on.

Test your sketch to see if the graphs are drawn when each button is clicked on by uncommenting the appropriate code in the sketch tab.

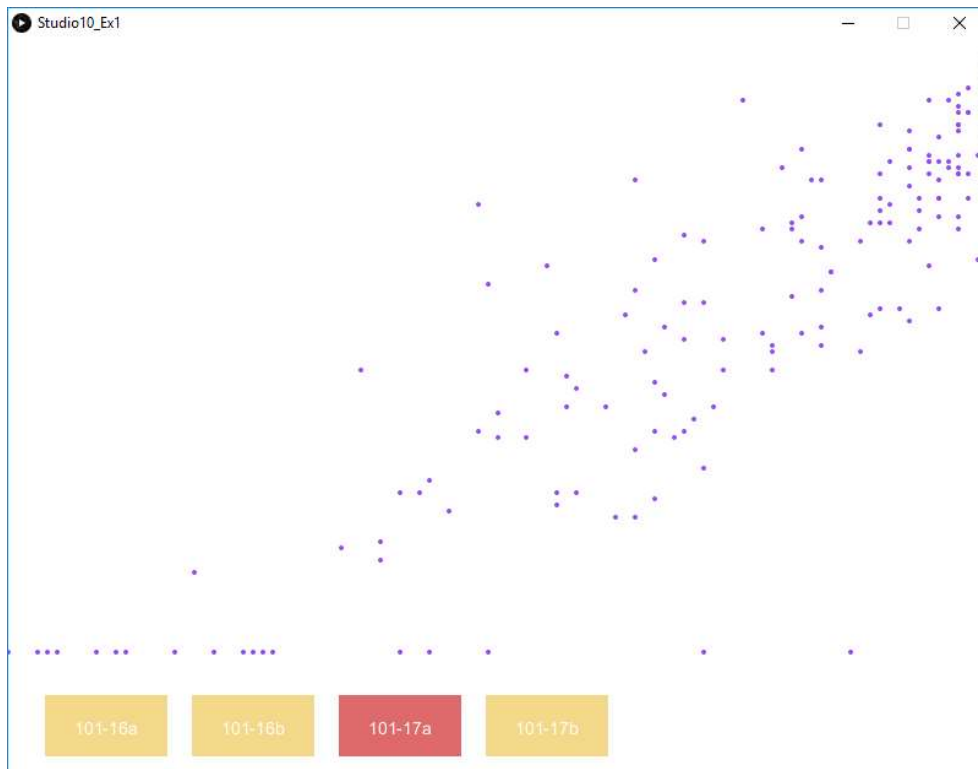
7. Modify your button class(es) so that when a button is selected it is drawn in a selected colour and when it is not selected it is drawn in its normal colour.



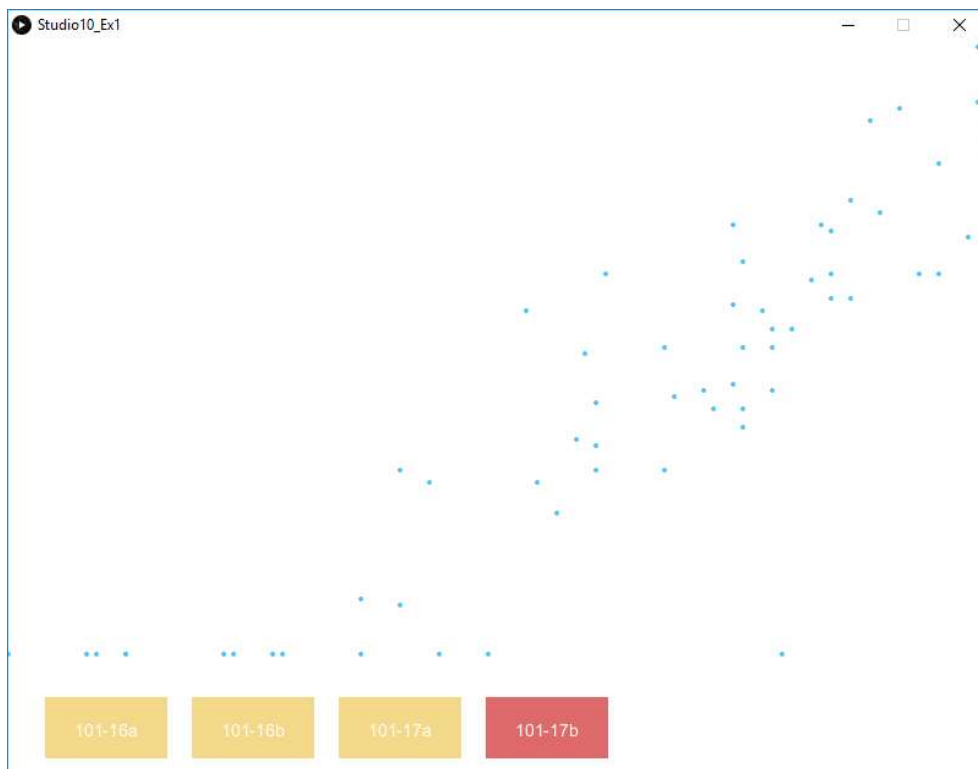
Screen shot of 101-16a



Screen shot of 101-16b



Screen shot of 101-17a



Screen shot of 101-17b